

Pronob Kumar Barman

[✉ Email](#) [☎ 443-931-5511](#) [in LinkedIn](#) [GH GitHub](#) [G Google Scholar](#) [🏠 Homepage](#)

EDUCATION

University of Maryland, Baltimore County

PhD in Information Systems, CGPA: 3.85/4.00

01/2022 - 05/2025 (Anticipated)

Research Focus: Generative Models, NLP, Recommendation Systems

University of Kentucky

MS in Statistics, CGPA: 3.85/4.00

08/2019 - 05/2021

Advanced Theoretical Focus: Probability Theory, Statistical Inference, Experimental Design

University of Dhaka

MS in Statistics, CGPA: 3.46/4.00

10/2015 - 01/2017

BS in Statistics, CGPA: 3.46/4.00

01/2011 - 08/2015

PROFESSIONAL EXPERIENCE

University of Maryland, Baltimore County

Teaching Assistant

01/2022 – Present

- Mentored graduate students in **Natural Language Processing (NLP)** and **machine learning**, focusing on foundational techniques and cutting-edge methods like **Transformer models (BERT, GPT)**, LSTM, and **deep neural networks**.
- Designed and delivered sessions on **large-scale data analysis**, incorporating **model optimization** and **deployment best practices** to solve real-world challenges.
- Collaborated with graduate students on integrating **state-of-the-art NLP techniques** into academic projects, fostering innovation and practical application.

Technuf LLC

Applied Scientist, Intern

06/2024 – 08/2024

- Engineered AI solutions by integrating **Large Language Models (LLMs)** such as GPT and BERT with **cybersecurity frameworks (SIEM and SOC systems)**, achieving a **25% improvement in threat detection accuracy**.
- Conducted advanced **trend analysis** on alert data, identifying patterns and optimizing detection strategies, reducing **false positives** by 20%.

Applied Scientist, Intern

08/2022 – 12/2022

- Streamlined data processing for **200K+ healthcare records** using optimized **SQL queries** in PostgreSQL, improving system efficiency by **40%**.
- Built interactive **Power BI dashboards** to visualize critical healthcare metrics, enabling **data-driven decision-making**.
- Automated testing workflows with **Flutter**, reducing debugging time and improving product release efficiency.

University of Kentucky

Teaching Assistant

08/2019 – 05/2021

- Conducted hands-on training in **statistical modeling**, **database querying (SQL)**, and **data analysis tools**, improving student proficiency in real-world problem-solving.
- Assisted in research projects requiring **quantitative analysis**, focusing on model evaluation metrics and statistical rigor.

Bangladesh Bank

Senior Data Scientist

04/2018 – 12/2021

- Designed and implemented advanced **machine learning models** to improve **credit risk assessment**, reducing default rates by **15%** and saving millions annually.
- Utilized statistical models to optimize **foreign direct investment (FDI)** strategy, leading to a **15% increase in inflows (\$500 million/year)**.
- Collaborated with **cross-functional teams** to develop **predictive analytics models**, improving loan portfolio performance and decision-making.

SKILLS

- **Programming Languages:** Python, R, Java, Matlab, SAS
- **Machine Learning Frameworks:** TensorFlow, PyTorch, scikit-learn, Keras
- **Data Science Libraries:** pandas, NumPy, SciPy, NetworkX, StatsModels
- **Generative AI:** GPT-3/4, LangChain, Hugging Face, OpenAI API, ChromaDB
- **Visualization Tools:** Tableau, Power BI, Matplotlib, Seaborn, Plotly
- **Natural Language Processing:** BERT, GPT, SpaCy, NLTK
- **Data Management:** PostgreSQL, Pinecone, ChromaDB
- **Research Tools:** NVivo, Qualtrics, LaTeX, VScode, Git, Shell (Bash), Jupyter Notebook, R Markdown
- **Soft Skills:** Leadership, Collaboration, Communication, Problem-Solving

Online Healthcare Support Group Formation and Recommendation System

- Developed two novel generative models: **Group-specific Dirichlet Multinomial Regression (gDMR)** and **Group-specific Structural Topic Model (gSTM)**, aimed at automating and personalizing support group formation in online healthcare platforms.
- The gDMR model extends traditional **Dirichlet Multinomial Regression (DMR)** by introducing group-specific parameters and leveraging user demographics, content, and interaction patterns, enabling the formation of semantically coherent and demographically relevant groups.
- The gSTM model enhances **Structural Topic Modeling (STM)** by incorporating group-specific parameters and node embeddings, improving group stability and enabling more personalized recommendations.
- Achieved a **60% improvement in topic coherence** and a **75% boost in recommendation accuracy**, validated through extensive real-world experiments.
- Released open-source implementations, including an [R package for gSTM](#) and a [Python package for gDMR](#), facilitating adoption in diverse research and application domains.

OTHER PROJECTS

LLM Generative AI with LangChain and OpenAI API [Github](#)

- Developed a **retrieval-augmented generation system** for advanced semantic search, using **Pinecone** and **ChromaDB**.

End-to-End Medical Chatbot with Llama2 and Hugging Face API [Github](#)

- Developed a **medical chatbot with Llama2**, improving **patient interaction flow by 30%** through **LangChain** enhancements.

Cognitive and Affective State Recognition in iVR Using Multimodal Signals [Github](#)

- Developed inverse inference models using **multimodal signals**, integrating neuro-physiological data such as EEG and eye-tracking to map cognitive and affective states in immersive virtual reality (iVR).

Predictive Policing with GAN

- Implemented a Generative Adversarial Network (GAN) model to identify and mitigate bias in predictive policing algorithms, achieving a 20% reduction in false-positive rates.

Classification of Medical Specialties from Transcriptions

- Achieved 98% classification accuracy by developing a SMOTE-enhanced machine learning model for classifying medical specialties from transcription data.

PUBLICATIONS

- Balogun, Z. A., **Barman, P. K.**, & Reynolds, T. L. (2025). "Cultivating a Personalized Digital Health Ecosystem: Insights from a Mixed Methods Survey of U.S. Adults." *ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)*, 2025. (Under Review)
- Walid, M. A. A., Devnath, M. K., Muiz, B., Melon, M. M. H., **Barman, P. K.**, & Barman, P. K. (2024). "Efficient Graph-Based Intrusion Detection for CAN Bus Security with Minimal Training Data." *Proceedings of the 6th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2024)*, 2024.
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium*, 2024.
- Faruq, M. O., Walid, M. A. A., Baowaly, M. K., Devnath, M. K., Ejaz, M. S., **Barman, P. K.**, & Sattar, A. S. (2024). "Shielding Privacy: A Technique of Extenuating Composition Attacks in Various Independent Data Publication." *Bulletin of Electrical Engineering and Informatics*, 13(4), 2677-2686.
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012*.

PEER REVIEWER

- NeurIPS 2024 Workshop on Behavioral ML
- AMIA 2024 Annual Symposium